

Lab: Laser Radar

LD14 is mainly composed of a laser ranging core, a wireless power transmission unit, a wireless communication unit, an angle measurement unit, a motor drive unit, and a mechanical housing. The LD14 ranging core adopts triangulation technology, which can perform 2300 ranging times per second. During each distance measurement, LD14 emits infrared laser from a fixed angle, which is reflected back to the receiving unit upon encountering the target object. Calculate the distance by establishing a triangular relationship between the laser, target object, and receiving unit. After obtaining distance data, LD14 will fuse the angle values measured by the angle measurement unit to form the point cloud number. According to the data, the point cloud data is then sent to an external interface through wireless communication. At the same time, the motor drive unit will drive the motor to the specified speed through PID closed-loop control algorithm.

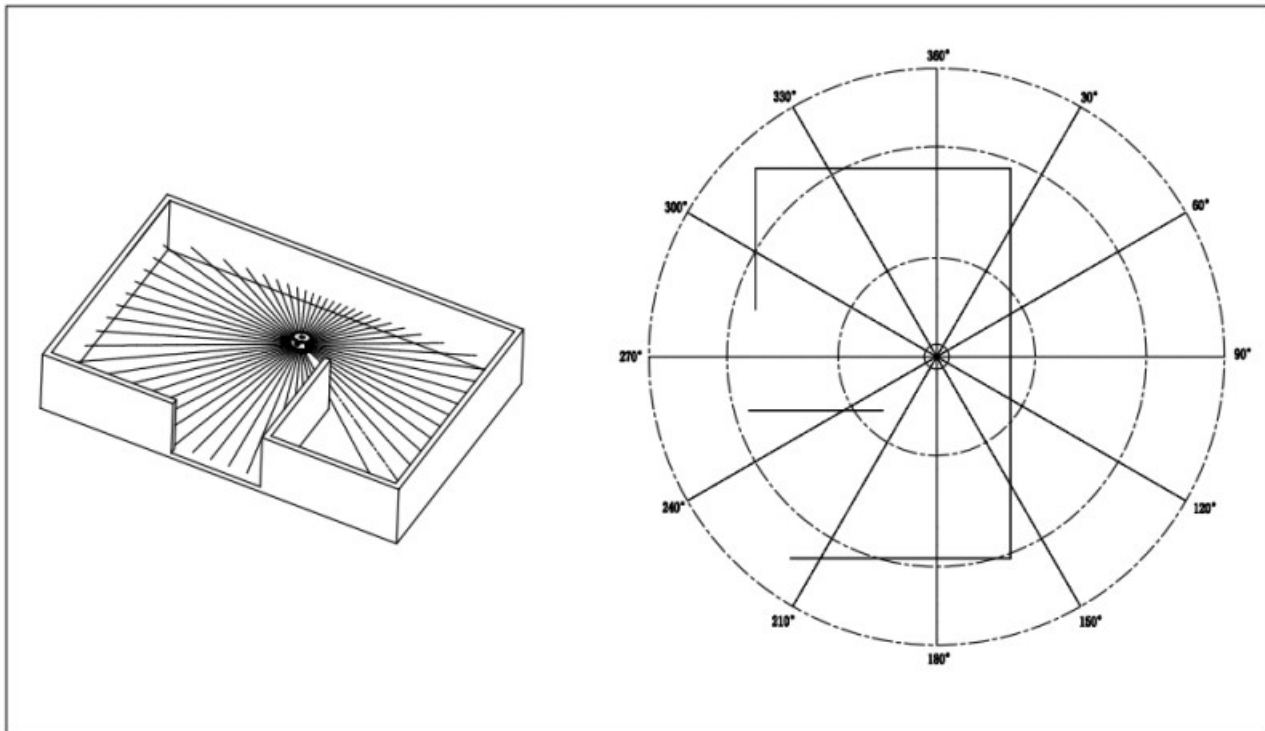
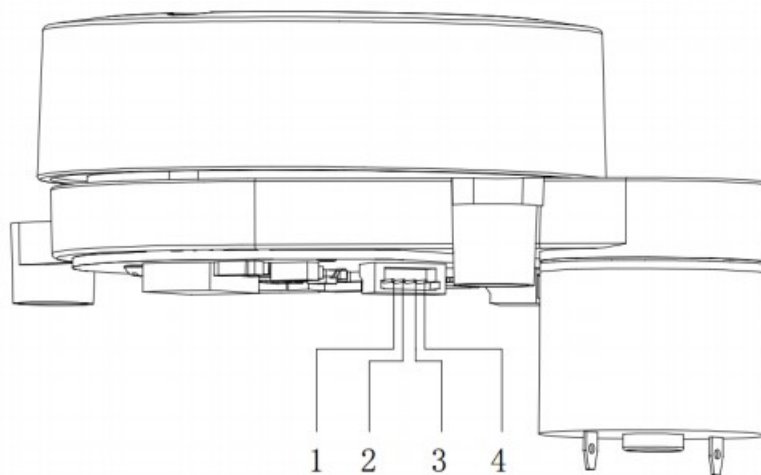


Fig.1 Environmental scanning image using LD14

1 LD14

The LD14 uses four PIN connectors to connect with external systems for power supply and data reception. The specific interface definition and parameter requirements are shown in the following figure.



序号	信号名	类型	描述	最小值	典型值	最大值
1	PWM	输入	电机控制信号	0V	-	3.3V
2	GND	供电	电源负极	-	0V	-
3	Tx	输出	雷达数据输出	0V	3.3V	3.5V
4	P5V	供电	电源正极	4.5V	5V	5.5V

Fig.2 The pin diagram of LD14

2 Test program

Students should learn how to use LD14 and obtain distance data around the car.

2.1 Experimental requirements

Without using PWM to control the speed of the LD14, students learn how to read the data output by the LD14. The data format is shown in Figure 3. Distinguish the direction closest to the car from the read data.

起始符	VerLen	雷达转速		起始角度		数据	结束角度		时间戳		CRC 校验
54H	2CH	LSB	MSB	LSB	MSB		LSB	MSB	LSB	MSB	1Byte



测量点 1			测量点 2			...	测量点 n		
距离值		信号强度	距离值		信号强度		距离值		信号强度
LSB	MSB	1 Byte	LSB	MSB	1 Byte	...	LSB	MSB	1 Byte

Fig.3 The data format

2.2 Experimental requirements

Students learn to control the rotation of LD14 using PWM waveform. Pay attention to the frequency and duty cycle of PWM. Use LD14 to detect the distance of the surrounding environment of the car, find the direction with the closest distance, rotate the car towards the direction with the closest distance, and retreat 10-20cm in the opposite direction. Continue the above operation twice.